ARAB ACADMY FOR SCIENCE & TECHNOLOGY
OGYCOLLEGE OF Engineering & Technology

Lecturer : Staff of Chemistry

Course : Eng. Chemistry

Course No. : BA118

Date : 2 / 6 / 2014

Marks: 40

Time: 09:00 – 11:00 (2 hrs)

Final Examination Paper

SECTION I {25 marks}

Answer the following questions

1. a- Explain the defects of Scale formation inside the Boilers.
   b- Explain haw to Removal of **Dissolved Oxygen** from feeding water of boiler (law and high pressure), state the advantages and disadvantages.
   c- Compare between scale and Sludge.

   {8marks}

2. a- Deduce the combustion equation of ***Methane gas*** (CH_4) complete combustion).
   b- Compare between combustion of **Carbon** (complete & incomplete)**and Hydrogen**.
   c- Explain the effects of the following upon property and the efficiency of fuel:
      (i) Increase or decrease the viscosity than the limited value.
      (ii) Sulphur content.
      (iii) Carbon – Hydrogen content .
      (iv) Surface tension.

   {8 marks}

3. a- Write in-details about **Corrosive environments.**
   b- State and explain **Reasons** of Pitting corrosion and the **Factors** affecting.
   c- Describe and explain **Impressed** Current method of Cathodic Protection and compare between Sacrificial and Impressed current methods.

   {9 marks}

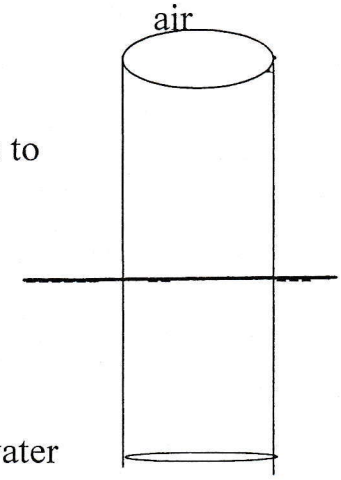
NB; atomic mass H=1,C=12, O=16, N=14, CV_C=33.7 &10.5, CV_H=65.4, CV_S=9.1 MJ. CV_{CH_4}=20MJ

SECTION II {15 Marks}

Encircle The Best Choice from a,b,c and of The Following

1. The mainly factors affecting atmospheric corrosion are
   - a- oxygen and humidity .
   - b- humidity and temperature.
   - c- dust and pressure.
   - d- all of the above.

2. Corrosion of metallic structure (as shown in Fig.) is due to
   - a- differential aeration corrosion.
   - b- pitting corrosion.
   - c- atmospheric corrosion.
   - d- all of the above.



3. Erosion corrosion increasing with
   - a- increasing the velocity of fluid.
   - b- presence of waves.
   - c- presence of suspended particles in the fluid.
   - d- all of the above.

4. To successful coating protection the following must be occurs
   - a- good cleaning the surface.
   - b- good applied method or good technique.
   - c- good barrier.
   - d- both a and b.

5. Reasons of Pitting corrosion are
   - a- the surface and the corrosive environment are non uniform.
   - b- potential difference.
   - c- localized impurities.
   - d- both a and c.

6. Galvanic corrosion takes place due to
   a- bimetallic couple.
   b- different metals or alloys.
   c- potential difference.
   d- both a and b.

7. The conditions of Electroplating technique are
   a- D.C power supply and electrolyte solution related to anode.
   b- coating metal must be connected with +ve terminal of power supply.
   c- both a and b.
   d- D.C power supply, connection and specific electrolyte solution.

8. Incomplete combustion of fuel is due to
   a- insufficient air supply and lower or higher viscosity .
   b- increase % of ash.
   c- formation of carbon monoxide.
   d- all of the above.

9. To burn 1 Kg. of Carbon (complete) ,needs to amount of air equals
   a- 11.594 Kg.
   b- 8/3 Kg.
   c- 4/3 Kg.
   d- non of the above.

10. The amount of energy which realize by 1 kg. fuel(composed of 40% C & 60% H_2 complete combustion) equals
   a- 98.8 MJ.
   b- 52.72 MJ .
   c- 43.44 MJ.
   d- none of the above.

11. Flash point of fuel may be indicating to
   a- probability of self ignition and condition of storage.
   b- calorific value.
   c- density of fuel.
   d- both a and c.

12. Hard water is owing to presence of
    a- calcium and magnesium salts.
    b- calcium and magnesium Cations.
    c- bicarbonate and sulphate salts of calcium and magnesium.
    d- both b and c.

13. pH value of water after Cation exchange resins equals
    a- more than 7.
    b- less than 7.
    c- equal 7.
    d- non of the above.

14. Cation exchange resin, can be activated or regenerated by washing the resin by
    a- acidic solution.
    b- diluted sodium hydroxide solution.
    c- diluted HCl solution.
    d- diluted HNO_3 solution.

15. To removal oils or grease from water can be used the following material
    a- aluminum sulphate.
    b- alum compounds.
    c- coagulating agent.
    d- gelatinous compound.

Good Luck

Members of Course Examination Committee	Signature	Date
Prof. Mohamed Amer- Dr. Ibrahim Hassan		
Dr. Ibrahim Hassan		